

27/04/2026

Assignment 2:

Task 1:

$$P = 100 \text{ W}$$

luminous efficacy 11 lm/W

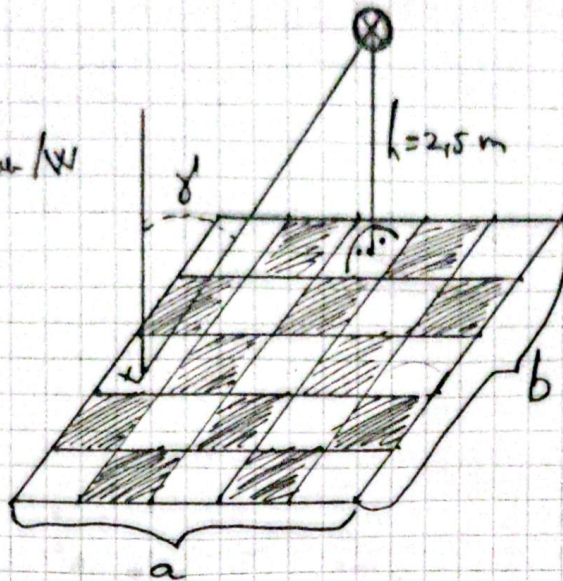
$$\eta = 0.8$$

$$h = 2.5 \text{ m}$$

$$a = b = 5 \text{ m}$$

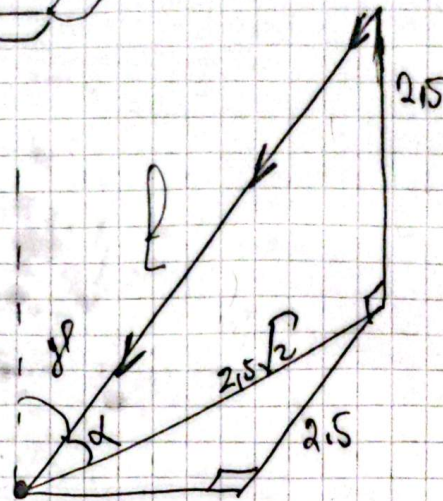
$$E(l_x) = ?$$

$$\gamma = ?$$



Solution:

$$l = \sqrt{2.5^2 + (2.5\sqrt{2})^2} = 3.354 \text{ m}$$



(3)

$$\Phi = P \cdot \eta_{\text{eff}} = 11 \cdot 100 = 1100 \text{ lm}$$

$$E = \frac{\Phi}{l^2} \cdot \eta = \frac{1100}{3.354^2} \cdot 0.8 = 78.227 \text{ lx}$$

$$\gamma = 90 - 35.26 = 54.74^\circ$$

$$l = \sqrt{2.5^2 + (2.5\sqrt{2})^2} = 3.354 \text{ m} \quad (1)$$

$$\tan \alpha = \frac{2.5}{2.5\sqrt{2}} = \frac{1}{\sqrt{2}} \quad \alpha = 35.26^\circ \quad \text{then} \quad (2)$$

$$\Phi = P \cdot \eta_{\text{eff}} = 100 \cdot 11 = 1100 \text{ lm}$$

$$E_t = \frac{\Phi}{l^2}$$

$$E_t = \frac{\Phi}{l^2} = \frac{1100}{3.354^2} = 97.784 \text{ lx}$$

$$E_u = E_t \cdot \eta$$

Task 2:

Components:

$$X=20 \quad Y=60 \quad Z=20$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 0,4185 & -0,1587 & -0,0828 \\ -0,0912 & 0,2524 & 0,0157 \\ 0,0009 & -0,0025 & 0,1786 \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

① Trichromatic Components:

$$R = (0,4185 \cdot 20) + (-0,1587 \cdot 60) + (-0,0828 \cdot 20) = -28,08$$

$$G = (-0,0912 \cdot 20) + (0,2524 \cdot 60) + (0,0157 \cdot 20) = 13,634$$

$$B = (0,0009 \cdot 20) + (-0,0025 \cdot 60) + (0,1786 \cdot 20) = 3,44$$

② Chromaticity Coordinates:

$$S = R + G + B = 14,266$$

$$r = \frac{R}{S} = \frac{-28,08}{14,266} = -0,1968$$

$$g = \frac{G}{S} = \frac{13,634}{14,266} = 0,9557$$

$$b = \frac{B}{S} = \frac{3,44}{14,266} = 0,2411$$